WORKSHEET 0.2

Trigonometry Without Hesitation

Unit 0: Readiness

Key ideas

- Calculus uses **radians**: π radians is $180°$.
- For an angle outside the first quadrant, use its **reference angle** and attach the sign: sine is positive in quadrants I and II, cosine in I and IV, tangent in I and III.
- $\sin^2 x + \cos^2 x = 1$, $\quad \sin 2x = 2\sin x \cos x$, $\quad \cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1$.
- To solve an equation, move everything to one side and **factor**. Never divide by $\sin x$ or $\cos x$.

Worked examples

Example 1 · Exact values

Find $\cos\dfrac{3\pi}{4}$, $\sin\dfrac{11\pi}{6}$ and $\tan\dfrac{7\pi}{6}$.

Solution. Reference angles $\dfrac{\pi}{4}$, $\dfrac{\pi}{6}$, $\dfrac{\pi}{6}$, in quadrants II, IV and III.

$-\dfrac{\sqrt{2}}{2}$, $\quad -\dfrac{1}{2}$ **and** $\quad \dfrac{1}{\sqrt{3}}$.

Example 2 · A linear equation

Solve $2\cos x + 1 = 0$ for $0 \le x < 2\pi$.

Solution. $\cos x = -\dfrac{1}{2}$. The reference angle is $\dfrac{\pi}{3}$ and cosine is negative in quadrants II and III.

$x = \dfrac{2\pi}{3}$ **and** $x = \dfrac{4\pi}{3}$.

Example 3 · Using an identity first

Solve $\cos 2x + \sin x = 0$ for $0 \le x < 2\pi$.

Solution. $1 - 2\sin^2 x + \sin x = 0$, so $2\sin^2 x - \sin x - 1 = (2\sin x + 1)(\sin x - 1) = 0$. $\sin x = 1$ gives $\dfrac{\pi}{2}$; $\sin x = -\dfrac{1}{2}$ gives $\dfrac{7\pi}{6}$ and $\dfrac{11\pi}{6}$.

$x = \dfrac{\pi}{2}$, $\dfrac{7\pi}{6}$, $\dfrac{11\pi}{6}$.

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Practice

CORE

1. Find $\sin\dfrac{3\pi}{2}$ and $\cos\dfrac{2\pi}{3}$.

2. Convert $\dfrac{5\pi}{4}$ radians to degrees.

3. Solve $\sin x = \dfrac{\sqrt{3}}{2}$ for $0 \le x < 2\pi$.

4. Solve $\tan x = -1$ for $0 \le x < 2\pi$.

EXAM LEVEL

5. Solve $4\cos^2 x = 3$ for $0 \le x < 2\pi$.

6. Given $\cos\theta = -\dfrac{2}{3}$ with θ in quadrant II, find $\sin\theta$ and $\tan\theta$.

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7. Show that $\tan x + \cot x = \sec x \csc x$.

8. Find $\arcsin \dfrac{\sqrt{3}}{2}$, $\arctan\left(-\sqrt{3}\right)$ and $\arccos 0$.

Question 9 · Free-response practice

Let $g(x) = \cos 2x + 3\cos x + 2$ for $0 \leq x < 2\pi$.

(a) Find the exact value of $g\left(\dfrac{\pi}{3}\right)$.

(b) Write $g(x)$ in terms of $\cos x$ only.

(c) Find all values of x in the interval at which $g(x) = 0$.

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Worked solutions

1. -1 and $-\dfrac{1}{2}$.

2. $225°$.

3. $x = \dfrac{\pi}{3}$ and $x = \dfrac{2\pi}{3}$.

4. $x = \dfrac{3\pi}{4}$ and $x = \dfrac{7\pi}{4}$.

5. $\cos x = \pm\dfrac{\sqrt{3}}{2}$: $x = \dfrac{\pi}{6},\ \dfrac{5\pi}{6},\ \dfrac{7\pi}{6},\ \dfrac{11\pi}{6}$.

6. $\sin^2\theta = 1 - \dfrac{4}{9} = \dfrac{5}{9}$, and sine is positive in quadrant II: $\sin\theta = \dfrac{\sqrt{5}}{3}$, $\tan\theta = -\dfrac{\sqrt{5}}{2}$.

7. $\dfrac{\sin x}{\cos x} + \dfrac{\cos x}{\sin x} = \dfrac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \dfrac{1}{\sin x \cos x} = \sec x \csc x$.

8. $\dfrac{\pi}{3},\ -\dfrac{\pi}{3}$ and $\dfrac{\pi}{2}$.

9. (a) $\cos\dfrac{2\pi}{3} + 3\cos\dfrac{\pi}{3} + 2 = -\dfrac{1}{2} + \dfrac{3}{2} + 2 = 3$.

   (b) $2\cos^2 x - 1 + 3\cos x + 2 = 2\cos^2 x + 3\cos x + 1$.

   (c) $(2\cos x + 1)(\cos x + 1) = 0$. $\cos x = -\dfrac{1}{2}$ gives $\dfrac{2\pi}{3}$ and $\dfrac{4\pi}{3}$; $\cos x = -1$ gives π.

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